

7 Finance

7.5 Payday Loans and Credit Cards

7.5.1 Credit Cards

Example 7.5.1 Credit cards sometimes compound interest daily. Suppose that you owe \$1000 on a credit card with a 25% APR. You make a payment of \$100 (after interest is compounded for the month). How much will you still owe the bank? (Assume each month has exactly 30 days)

Use the compound interest formula, where the exponent is just 30. This might take some explaining.

Example 7.5.2 Suppose that you owe \$1000 on a credit card with 28% APR, and the minimum payment each month is \$50, additionally assume that you make no additional purchases (not a very realistic assumption, I know, but let's make this simple for ourselves). How much will we owe the bank after 3 months?

Example 7.5.3 Let's go the computer for this one. In the above situation from the previous example, how long will it take us to pay off the bank? How much do we pay?

Example 7.5.4 Some credit cards use an averaging process to calculate the interest owed. Let's assume that the following is true:

At beginning of month	You owe \$1000
On the 10th	you make a \$250 purchase
On the 15th	you make a \$65 purchase

How much interest will you be charged at the end of the month (assuming the card issuer has no grace period) on a credit card with a 25% APR? (Assume the month has 31 days)

1. Calculate daily balance

2. Calculate daily interest rate

3. Calculate interest

$DPR * \text{days} * \text{average balance}$

7.5.2 Payday Loans

Example 7.5.5 Payday loans are short-term loans where the interest is calculated up front. Consider the following example. You need \$400 dollars to make rent, but don't get paid for another two weeks. You decide to take out a payday loan, and the lender charges \$15 dollars interest for every \$100 dollars borrowed. How much interest will you pay if you pay back the loan on time? What is the APR of this loan?

Use the formula

$$A = P_0 \left(1 + \frac{r}{k} \right)$$

since interest is only compounded 1 time, and $k = 26$, since interest is compounded for a period of two weeks in this example. The APR is 390%.

7.5.3 Review of Finance

Formulas so far:

For these formulas, we have the following: P_0 is the principal. r is the interest rate. t is the number of time intervals (specified by problem) which have passed. A is the amount paid back. I is the interest.

Simple Interest	$A = P_0(1 + r)$
Simple Interest over Time	$A = P_0(1 + rt)$

For these formulas, we have the following: P_0 is the principal, or amount of a loan. r is the annual interest rate. t is the number of years which have passed. A is the total amount. k is the number of times interest is compounded per year. d is the amount we deposit (or withdraw) from the account regularly, or the payment amount on a loan.

Compound Interest	$A = P_0 \left(1 + \frac{r}{k}\right)^{kt}$
Savings Annuity	$A = d \frac{\left(1 + \frac{r}{k}\right)^{kt} - 1}{\frac{r}{k}}$
Payout Annuity/Loans	$P_0 = d \frac{1 - \left(1 + \frac{r}{k}\right)^{-kt}}{\frac{r}{k}}$

Example 7.5.6 Suppose you obtain a \$3,000 T-note with a 3% annual rate, paid quarterly, with maturity in 5 years. How much interest will you earn?

Example 7.5.7 You deposit \$300 in an account earning 5% interest compounded annually. How much will you have in the account in 10 years?

Example 7.5.8 You want to be able to withdraw \$30,000 each year for 25 years. Your account earns 8% interest.

1. How much do you need in your account at the beginning
2. How much total money will you pull out of the account?
3. How much of that money is interest?

Example 7.5.9 You want to buy a \$25,000 car. The company is offering a 2% interest rate for 48 months (4 years). What will your monthly payments be?

Example 7.5.10 A friend lends you \$200 for a week, which you agree to repay with 5% one-time interest. How much will you have to repay?

Example 7.5.11 You wish to have \$3000 in 2 years to buy a fancy new stereo system. How much should you deposit each quarter into an account paying 8% compounded quarterly?

Example 7.5.12 You can afford a \$700 per month mortgage payment. You've found a 30 year loan at 5% interest.

1. How big of a loan can you afford?
2. How much total money will you pay the loan company?
3. How much of that money is interest?

Example 7.5.13 You deposit \$1000 each year into an account earning 8% compounded annually.

1. How much will you have in the account in 10 years?
2. How much total money will you put into the account?
3. How much total interest will you earn?